Permutation-symmetric lattice tiling bodies require surface area at least n over square root log n

Mark Braverman, Dor Minzer

doi:10.4230/LIPIcs.CCC.2021.5; CC BY 4.0

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Editorial note. Transcription correction against officialPDFpage11: published membership j in K_i is rendered with the in macro; nativev1 had an invalid math-mode dotless-i command. Original formal author expression retained, raw native unchanged.

Allowed standard prerequisites (editorial).

- Gaussian estimates, elementary concentration, geometry of surface area and noise sensitivity. Full original source-local geometric/concentration statements and proofs included; cited standard background not recursively reproved

1 Introduction

A body $D \subseteq \mathbb{R}^n$ is said to be tiling the Euclidean space $\mathbb{R}^n$, if its translations by $\mathbb{Z}^n$ cover the entire space and have disjoint interiors. The foam problem asks for the least surface area a tiling body D can have. The problem had been considered by mathematicians already in the 19th century [33], and it also appears in chemistry, physics and engineering [30]. More recently, the problem had received significant attention in the theoretical computer science community due to its strong relation with the parallel repetition problem [15, 24, 2].

The simplest example for a body that tiles the Euclidean space is the solid cube, $D = [0, 1]^n$, which has surface area $2n$. At first glance, one may expect the solid cube to be the best example there is, or more modestly that any tiling body would need to have surface area $\Omega(n)$. The main results of [24, 2] show that this initial intuition is completely false, and that there are far more efficient

tiling bodies whose surface area is $O(\sqrt{n})$. This is surprising, since spheres — which are the minimizers of surface area among all bodies with a given, fixed volume (in this case volume 1), have $\Theta(\sqrt{n})$ surface area and seem to be very far from forming a tiling of $\mathbb{R}^n$. As we will shortly discuss, the existence of such surprising tiling body is intimately related to the existence of another surprising object – namely non-trivial strategies for 2-prover-1-round games, repeated in parallel. The main goal of this paper is to understand the symmetric variant of the foam problem, which is closely related to the symmetric variant of parallel repetition.

1.3 Our results

In this paper, our main object of study mainly are tilings of $\mathbb{R}^n$ using a *symmetric body*.

Definition 4. *A set $D \subseteq \mathbb{R}^n$ is called symmetric if for any $\pi \in S_n$ and $x \in \mathbb{R}^n$, it holds that $x \in D$ if and only if $\pi(x) \in D$.*

The main question we consider, is what is the least surface area a symmetric tiling body can have. Again, one has the trivial example of the solid cube $D = [0,1]^n$, but inspired by the non-symmetric variant of the problem, one may expect there to be better examples. We first show that while this is possible, the savings are much milder, and can be at most a multiplicative factor of $\sqrt{\log n}$.

Theorem 5. *Any symmetric tiling body D of volume 1 with piecewise smooth surface has surface area at least $\Omega\left(\frac{n}{\sqrt{\log n}}\right)$.*

Besides the quantitative result itself, we believe the argument used in the proof of Theorem 5 carries with it a lot of intuition regarding the additional challenge that the symmetric variants of the foam problem and the parallel repetition posses, and we hope that this intuition will help us to develop better understanding of symmetric parallel repetition in general. We remark that our proof actually shows a lower bound on the "noise sensitivity" parameter of the body, which is known to be smaller than the surface area of the body.

We complement Theorem 5 with a randomized construction showing that $O(\sqrt{\log n})$ savings are indeed possible.

Theorem 6. *There exists a symmetric tiling body D of volume 1 with piecewise smooth surface that has surface area $O\left(\frac{n}{\sqrt{\log n}}\right)$.*

2 Preliminaries

Notations. We write $X \lesssim Y$ or $X = O(Y)$ to say that there exists an absolute constant $C > 0$ such that $X \leqslant C \cdot Y$, and similarly write $X \gtrsim Y$ or $X = \Omega(Y)$ to say that there exists an absolute constant $c > 0$ such that $X \geqslant c \cdot Y$. We write $X \asymp Y$ or $X = \Theta(Y)$ to say that $Y \lesssim X \lesssim Y$.

We denote random variables by boldface letters such as $\mathbf{x}$ and $\mathbf{\Delta}$. We denote by $\mathcal{N}(\mu, \sigma^2)$ the distribution of a standard Gaussian random variable with mean μ and variance σ^2, and by $\mathcal{N}(\vec{\mu}, \Sigma)$ the distribution of a multi-dimensional Gaussian random variable with means $\vec{\mu}$ and covariance matrix Σ.

2.1 Needles

Definition 9. *Let $\delta > 0$, and let $a \in \mathbb{R}^n$. A random δ-needle is a line $\ell_{a,\mathbf{u}} = \{a + t \cdot \mathbf{u} \mid t \in [0,1]\}$ where the direction vector $\mathbf{u}$ is a chosen as a standard Gaussian $\mathcal{N}(0, \delta I_n)$.*

Given a tiling body D, a random δ-needle from D is a random δ-needle $\ell_{\mathbf{a},\mathbf{u}}$ where $\mathbf{a} \in D$ is chosen uniformly. Random needles are a useful tool to measure the surface area of a D, as shown in the following two lemmas. First, given a tiling body D and a needle $\ell_{a,u}$, we may think of the needle as "wrapping around" around D, i.e. its points are taken modulo D. We denote this "wrapped around" line by $\tilde{\ell}_{a,u}$. We will use the following formula from [32]; the case $n = 2$ is formula (8.10) therein, and the extension to general n is discussed in page 274.

Lemma 10. *There is a constant $C_n = \Theta(1)$, such that the following holds. Let S be a piecewise smooth surface in a tiling body D of volume 1, and let $\delta > 0$. Then*

$$\mathop{\mathbb{E}}_{\mathbf{a}\in D, \mathbf{u}\sim\mathcal{N}(0,\delta I_n)} \left[\left|\tilde{\ell}_{\mathbf{a},\mathbf{u}} \cap S\right|\right] = C_n \cdot \sqrt{\delta} \cdot \mathsf{area}(S).$$

Lemma 11. *Let D be a tiling body of volume 1, and let $\delta > 0$. Then*

$$\Pr_{\mathbf{a}\in D, \mathbf{u}\sim\mathcal{N}(0,\delta I_n)} [\ell_{\mathbf{a},\mathbf{u}} \cap \partial D \neq \emptyset] \leqslant \Theta(\sqrt{\delta})\mathsf{area}(\partial D).$$

Proof. Set $S = \partial D$, and note that whenever $\ell_{a,\delta u} \cap \partial D \neq \emptyset$, we have that $\left|\tilde{\ell}_{a,\delta u} \cap S\right| \geqslant 1$. Hence by the previous lemma we get that

$$\Pr_{\mathbf{a}\in D, \mathbf{u}\sim\mathcal{N}(0,\delta I_n)} [\ell_{\mathbf{a},\mathbf{u}} \cap \partial D \neq \emptyset] \leqslant \mathop{\mathbb{E}}_{\mathbf{a}\in D, \mathbf{u}\sim\mathcal{N}(0,\delta I_n)} \left[\left|\tilde{\ell}_{\mathbf{a},\mathbf{u}} \cap \partial D\right|\right] \leqslant \Theta(\sqrt{\delta})\cdot\mathsf{area}(\partial D).$$

□

We will use the above lemma to prove lower bounds on the surface area of a tiling body, by finding δ such that the probability on the left hand side of Lemma 11 is at least $\Omega(1)$; this would imply that $\mathsf{area}(\partial D) \geqslant \Omega(1/\sqrt{\delta})$.

2.2 Basic useful properties of tiling bodies

Lemma 12. *Let $D \subseteq \mathbb{R}^n$ be a symmetric body, such that for all $z \in \mathbb{Z}^n \setminus \{0\}$ we have $D \cap (D+z) = \emptyset$, and let $x \in D$. Then for every $1 \leqslant i, j \leqslant n$, if $x_i - x_j \in \mathbb{Z}$, then $x_i = x_j$.*

Proof. Assume towards contradiction $x_i - x_j$ is a non-zero integer k, and let $S_{i,j} \in S_n$ be the permutation that maps i to j, j to i and has any $r \neq i, j$ as a fixed point. Since D is symmetric, we have that $S_{i,j}(x) \in D$. Also, we have

$$x - S_{i,j}(x) = (x_i - x_j)(e_i - e_j) = k(e_i - e_j),$$

where e_i is the ith element in the standard basis. In other words, we get that $x = S_{i,j}(x) + z$ for non-zero $z \in \mathbb{Z}^n$, and therefore $x \in D + z$. This contradict the fact that D and $D + z$ are disjoint. □

Lemma 13. *Let D be a volume 1 tiling body, and choose $a = (a_1, \ldots, a_n) \in D$ uniformly at random. Then the random variable $(a_1 \pmod 1, \ldots, a_n \pmod 1)$ is uniform over $[0,1)^n$.*

Proof. Sample $\mathbf{x} \in [0,1)^n$, and take $\mathbf{a} = \mathbf{x} \pmod D$. Note that the distribution of $\mathbf{a}$ is uniform over D. Indeed, for that we note that the map $x \to x \pmod D$ is bijection from $[0,1)^n$ to D: otherwise, there were $x \neq x'$ in $[0,1)^n$ that are equal mod D, and therefore differ by non-zero lattice point (which is clearly impossible). Now as the distribution of $\mathbf{a} \pmod 1$ is just $\mathbf{x}$, the claim follows. □

3 The lower bound: proof of Theorem 5

In this section, we prove the lower bound on the surface area of a symmetric tiling body D. Throughout, we will have two parameters: σ, which is magnitude of each coordinates in the needle we consider (which will be of order $\frac{\sqrt{\log n}}{n}$), and an auxiliary parameter Z (which will be of order $\frac{n}{\log n}$). Let D be a symmetric tiling body containing 0. We denote by $\mathbf{a}$ a random point in D, and by $\mathbf{u}$ a Gaussian vector $\mathcal{N}(0, \sigma^2 I_n)$. We will prove that $\Pr_{\mathbf{a},\mathbf{u}}[\ell_{\mathbf{a},\mathbf{u}} \not\subseteq D] = \Omega(1)$, which by Lemma 11 implies that $\mathsf{area}(\partial D) \geqslant \Omega(1/\sigma)$. As $\sigma = \Theta(\sqrt{\log n}/n)$, this would establish Theorem 5.

Notations. For $x, y \in \mathbb{R}$, define

$$d(x,y) := \min_{z \in \mathbb{Z}, z \neq 0} |(x+z) - y| \in [0,1].$$

To gain some intuition for the definition of $d(x,y)$, suppose x and y are two entries of a point $a \in D$. Clearly, if $d(x,y)$ is small, then x, y nearly differ by an integer $z \neq 0$, and this says that the point a is somewhat close to the boundary of D (in the sense that Lemma 12 could kick in if we move along a direction that decreases this distance).

Our argument will indeed inspect $d(a_i, a_j)$ for all distinct $i, j \in [n]$ and the way they change along a random direction. A key measure that we will keep track of is the energy of a point $a \in D$, defined by

$$\Psi(a) := \sum_{i<j} e^{-Z \cdot d(a_i, a_j)}.$$

We show that for $\mathbf{a} \in_R D$ and $\mathbf{u} \sim \mathcal{N}(0, \sigma^2 I_n)$, if $\ell_{\mathbf{a},\mathbf{u}} \subseteq D$ with probability close to 1, then the energy of $\mathbf{a}$ increases along the line $\ell_{\mathbf{a},\mathbf{u}}$ with high probability, and in particular that $\Psi(\mathbf{a}+\mathbf{u}) > \Psi(\mathbf{a})$. We then argue that with high probability, this should be the case for the point $\mathbf{a}$ as well as for $\mathbf{a}-\mathbf{u}$, hence $\Psi(\mathbf{a}+\mathbf{u}) > \Psi(\mathbf{a}-\mathbf{u})$ with high probability. This event however can happen with probability at most 0.5 by symmetry, hence completing the proof.

3.1 Analyzing the energy along a random line

By definition of $d(x,y)$, we either have $d(x,y) = (x+z-y)$ or $d(x,y) = -(x+z-y)$ for some $z \in \mathbb{Z} \setminus \{0\}$, and this sign determines whether x, y need

to move in different directions or the same direction for $d(x,y)$ to get smaller. To capture this, we denote

$$\gamma(x,y) := \begin{cases} +1 & \text{if } d(x,y) = x + z - y \text{ for some } z \in \mathbb{Z},\ z \neq 0, \\ -1 & \text{otherwise.} \end{cases}$$

Next, we discuss the *energy* of a configuration, which is the key concept used in the proof. Let Z be a parameter to be chosen later (of the order $n/\log n$). As stated earlier, our goal is to analyze the behaviour of $\Psi(a)$ along a random σ^2-needle from a in direction u. Towards this end, note that we expect (at least if u_i, u_j are small) that $d(a_i + u_i, a_j + u_j) = d(a_i, a_j) + \gamma(a_i, a_j)(u_j - u_j)$, hence expect $\Psi(a+u)$ to be close to

$$\Psi(a,u) := \sum_{i<j} e^{-Z \cdot (d(a_i,a_j) + \gamma(a_i,a_j) \cdot (u_i - u_j))}.$$

Indeed, this is the content of the following claim.

Claim 14. *Suppose $|u_i| \leqslant 1/20$ for all i, and $a + [0,1] \cdot u \subset D$, then*

$$|\Psi(a+u) - \Psi(a,u)| \leqslant n^2 \cdot e^{-Z/4}.$$

Proof. We consider the contribution of each pair (i,j) to $\Psi(a+u)$ and $\Psi(a,u)$ separately. Without loss of generality we may only consider pairs i,j that $\gamma(a_i,a_j) = 1$, and thus $d(a_i,a_j) = a_i - a_j + z$ for some $z \in \mathbb{Z}$, $z \neq 0$. Let

$$d = a_i - a_j + z + (u_i - u_j) = (a_i + u_i) - (a_j + u_j) + z.$$

First, we argue that $d \geqslant 0$. Otherwise, since $a_i - a_j + z \geqslant 0$ it follows by continuity that there is $\lambda \in [0,1]$ such that $a_i - a_j + z + \lambda(u_i - u_j) = 0$, and hence the point $a + \lambda u$ has entries that differ by an integer $z \neq 0$, and this contradicts Lemma 12 (as $a + \lambda u \in D$). We now consider two cases:

- **Case 1**: $d \in [0, 0.5]$. In this case, we have $d(a_i + u_i, a_j + u_j) = d$, and thus the contribution of the pair (i,j) to both sums is the same ($e^{-Z \cdot d}$).
- **Case 2**: $d > 0.5$. Since $|u_i - u_j| \leqslant 0.1$, it follows that $d(a_i, a_j) = d - (u_i - u_j) > 0.4$, which implies $d(a_i + u_i, a_j + u_j) > 0.3$. Therefore, the contribution to $\Psi(a,u)$ from i,j is at most $e^{-0.4 \cdot Z}$ and to $\Psi(a+u)$ is at most $e^{-0.3 \cdot Z}$, and in particular (i,j) contributes (in absolute value) at most $e^{-Z/4}$ to the difference between the sums.

Taking a sum over all pairs (i,j) concludes the proof. □

3.2 Analyzing the expectation and variance of $\Psi(a, \mathbf{u})$

Next, we consider $\Psi(a, \mathbf{u})$ as a random variable over the choice of $\mathbf{u}$ and compute its expectation and variance. In both computations we will use the well-known fact that $\mathbb{E}[e^{-Z \cdot N(0,c^2)}] = e^{Z^2 c^2/2}$ for all $c > 0$.

Claim 15. *For every $a \in \mathbb{R}^n$ we have $\mathbb{E}_{\mathbf{u} \sim \mathcal{N}(0, \sigma^2 I_n)} [\Psi(a, \mathbf{u})] = \Psi(a) \cdot e^{(Z \cdot \sigma)^2}$.*

Proof. By linearity of expectation we have that

$$\mathop{\mathbb{E}}_{\mathbf{u}\sim\mathcal{N}(0,\sigma^2 I_n)}[\Psi(a,\mathbf{u})] = \sum_{i<j} e^{-Z\cdot d(a_i,a_j)} \cdot \mathop{\mathbb{E}}_{\mathbf{u}\sim\mathcal{N}(0,\sigma^2 I_n)}\left[e^{-Z\cdot\gamma(a_i,a_j)\cdot(\mathbf{u}_i-\mathbf{u}_j)}\right].$$

Note that the above expectation does not depend on i,j: for every i,j the distribution of $\mathbf{u}_i - \mathbf{u}_j$ is $N(0,\sigma^2) - N(0,\sigma^2) \sim N(0,2\sigma^2)$, so it is symmetric around 0 and thus the sign $\gamma(a_i,a_j)$ does not affect the expectation. Hence we have

$$\mathop{\mathbb{E}}_{\mathbf{u}\sim\mathcal{N}(0,\sigma^2 I_n)}[\Psi(a,\mathbf{u})] = \Psi(a)\cdot\mathbb{E}[e^{Z\cdot N(0,2\sigma^2)}] = \Psi(a)\cdot e^{Z^2\sigma^2}. \qquad \square$$

Next, we turn our attention into upper bounding the variance of $\Psi(a,\mathbf{u})$, and for that we first define the notion of *good* points $a \in D$ and prove two preliminary claims. We say a point a is *good* if any interval of length $(10\log n)/n$ on the torus contains at least $\log n$ and at most $100\log n$ coordinates from $a (\mathrm{mod}\ 1)$. Note by Lemma 13, if $\mathbf{a}$ is chosen randomly from D then $\mathbf{a}$ (mod 1) is uniform over $[0,1)^n$ and by Chernoff bound is easily shown to be good with probability > 0.999.

We first show that good points have high energy.

Claim 16. *There exists $c_2 > 0$, such that for $Z = 0.1\frac{\log n}{n}$, if a is good then $\Psi(a) > c_2 \log^2 n$.*

Proof. Partition the torus $[0,1)$ into $m = n/(10\log n)$ disjoint intervals of length $1/m = (10\log n)/n$ each. We say that I_i is *unanimous*, if there is $b_i \in \mathbb{R}$ (called anchor) such that (1) $b_i (\mathrm{mod}\ 1)$ is the middle of I_i, and (2) for the majority of points $a_j \in I_i$, $|a_j - b_i| < 1/m$.

We consider two cases:

Case 1: There is an interval I_i that is not unanimous. Note that there are at least $\log n$ coordinates j of a such that $a_j \in I_i$. Let $j^\star$ be such coordinate, and write $a_{j^\star} = z_{j^\star} + \{a_{j^\star}\}$ where $z_{j^\star} \in \mathbb{Z}$ and $\{a_{j^\star}\}$ is the fractional part of $a_{j^\star}$. Consider $b = z_{j^\star} + m_i$ where m_i is the middle of I_i. Then since I_i is not unanimous, b is not an anchor of it and so there are at least $\frac{1}{2}\log n$ coordinates of a, say $(a_k)_{k\in K_{i,j^\star}}$ that mod 1 are in I_i, and $|a_k - b| \geqslant 1/m$. Writing $a_k = z_k + \{a_k\}$, we observe that $z_k \neq z_{j^\star}$, since otherwise $|a_k - b| = |\{a_k\} - m_i| \leqslant 1/(2m)$. Hence the difference $a_k - a_{j^\star}$ is $10\log n/n$ close to an integer $z_k - z_{j^\star} \neq 0$, and so $d(a_k, a_{j^\star}) \leqslant 10\log n/n$, and the contribution of $\Psi(a)$ is at least e^{-1}. Summing we get

$$\Psi(a) \geqslant \frac{1}{2}\sum_{j^\star : a_{j^\star}\in I_i}\ \sum_{k\in K_{i,j^\star}} e^{-Zd(a_k,a_{j^\star})} \geqslant \frac{1}{2}\sum_{j:a_j\in I_i} e^{-1}\,|K_{i,j^\star}| \geqslant \frac{1}{4e}\log^2 n.$$

Case 2: All intervals are unanimous. Let b_i be an anchor of I_i. Note that since the fractional part of two adjacent anchors, i.e. of b_i, b_{i+1}, are $1/m$ apart, we have that either $|b_i - b_{i+1}| \leqslant 1/m$ or $|b_i - b_{i+1}| \geqslant 1 - 1/m$. We claim there exists i for which the latter condition holds. To see this, assume that for all $i = 1,\ldots,m-1$ we have that the first condition holds. Then we have $b_i = z + i\frac{10\log n}{n}$ for some $z \in \mathbb{Z}$ for all $i = 1,\ldots,m$, and hence $|b_m - b_1| \geqslant 1 - 1/m$ (and the condition holds for $i = m$).

Thus, we fix i such that $|b_i - b_{i+1}| \geqslant 1 - 1/m$, and thus $b_i - b_{i+1} = z + \alpha$ for $z \neq 0$ and $|\alpha| \leqslant 1/m$. Let K_i be the coordinates j of a such that $|a_j - b_i| \leqslant 1/m$

for $j \in K_i$ and similarly define K_{i+1}. We have that $a_r - a_j = z + \alpha + (a_r - b_{i+1}) + (a_j - b_i)$, hence $a_r - a_j = z + \beta$ for $|\beta| \leqslant 3/m$ for all $r \in K_{i+1}$, $j \in K_i$. Thus $d(a_r, a_j) \leqslant 3/m$, and we get

$$\Psi(a) \geqslant |K_i|\,|K_{i+1}|\,e^{-Z\cdot 3/m} \geqslant \frac{1}{4}e^{-3}\log^2 n \qquad \square$$

Let $C_i = \sum_{j\neq i} e^{-Z\cdot d(a_i,a_j)}$ be the contribution of a_i to $\Psi(a)$. Note that $\Psi(a) = \frac{1}{2}\sum_i C_i$.

Claim 17. *There exists $c_3 > 0$, such that if a is good, then for all i we have $C_i < c_3\Psi(a)/\log n$.*

Proof. Note that $d(a_i, a_j) \geqslant |\{a_i\} - \{a_j\}|$. Since any interval of length $10\log n/n$ on the torus contains at most $100\log n$ points of a, we have that the number of j's such that $|\{a_i\} - \{a_j\}|$ is between $10\log n/n \cdot k$ and $10\log n/n \cdot (k+1)$ is at most $200\log n$ (for all k). Therefore,

$$C_i < 200\log n \cdot \sum_{k=0}^{\infty} e^{-Z\cdot k\cdot(10\log n)/n} = 200\log n \cdot \sum_{k=0}^{\infty} e^{-k} \leqslant 400\log n.$$

Using Claim 16, we may bound $\log n \leqslant \frac{1}{c_2}\frac{\Psi(a)}{\log n}$, finishing the proof. $\square$

We are now ready to bound the variance of $\Psi(a, \mathbf{u})$.

Claim 18. *There exists $c_1 > 0$ such that the following holds. Let $Z = n/10\log n$, let $a \in \mathbb{R}^n$ be good and let $\mathbf{u} \sim \mathcal{N}(0, \sigma^2 I_n)$. Then*

$$\mathsf{var}_{\mathbf{u}}[\Psi(a, \mathbf{u})] \leqslant \frac{c_1}{\log n} \cdot (e^{4(Z\cdot\sigma)^2} - e^{2(Z\cdot\sigma)^2}) \cdot \Psi(a)^2.$$

Proof. Using Claim 15 to compute the expectation of $\Psi(a, \mathbf{u})$, we have by definition that

$$\mathsf{var}_{\mathbf{u}}(\Psi(a,\mathbf{u})) = \mathop{\mathbb{E}}_{\mathbf{u}}\left[\left(\sum_{i<j} e^{-Z\cdot d(a_i,a_j)} \cdot (e^{Z\cdot\gamma(a_i,a_j)\cdot(\mathbf{u}_i-\mathbf{u}_j)} - e^{(Z\cdot\sigma)^2})\right)^2\right]$$

$$= \sum_{i<j} e^{-2Z\cdot d(a_i,a_j)} \cdot \mathop{\mathbb{E}}_{\mathbf{u}}\left[\left(e^{Z\cdot\gamma(a_i,a_j)\cdot(\mathbf{u}_i-\mathbf{u}_j)} - e^{(Z\cdot\sigma)^2}\right)^2\right]$$

$$+ \sum_{\substack{(i,j,k)\\ \text{distinct}}} e^{-Z\cdot(d(a_i,a_j)+d(a_i,a_k))} \cdot \mathop{\mathbb{E}}_{\mathbf{u}}\left[(e^{Z\cdot\gamma(a_i,a_j)\cdot(\mathbf{u}_i-\mathbf{u}_j)} - e^{(Z\cdot\sigma)^2})(e^{Z\cdot\gamma(a_i,a_k)\cdot(\mathbf{u}_i-\mathbf{u}_k)} - e^{(Z\cdot\sigma)^2})\right].$$

Here, we used that fact that if i, j, k, r are all distinct then $e^{Z\cdot\gamma(a_i,a_j)\cdot(\mathbf{u}_i-\mathbf{u}_j)}$, $e^{Z\cdot\gamma(a_k,a_r)\cdot(\mathbf{u}_k-\mathbf{u}_r)}$ are independent with expectation $e^{(Z\cdot\sigma)^2}$, hence the contribution of these terms is 0. Computing, we see that

$$\mathop{\mathbb{E}}_{\mathbf{u}}\left[\left(e^{Z\cdot\gamma(a_i,a_j)\cdot(\mathbf{u}_i-\mathbf{u}_j)} - e^{(Z\cdot\sigma)^2}\right)^2\right] = \mathbb{E}\left[e^{Z\cdot N(0,8\sigma^2)}\right] - e^{2(Z\cdot\sigma)^2} = e^{4(Z\cdot\sigma)^2} - e^{2(Z\cdot\sigma)^2},$$

and

$$\begin{aligned}
&\mathop{\mathbb{E}}_{\mathbf{u}}\left[(e^{Z\cdot\gamma(a_i,a_j)\cdot(\mathbf{u}_i-\mathbf{u}_j)}-e^{(Z\cdot\sigma)^2})(e^{Z\cdot\gamma(a_i,a_k)\cdot(\mathbf{u}_i-\mathbf{u}_k)}-e^{(Z\cdot\sigma)^2})\right]\\
&=\mathbb{E}\left[e^{(\gamma(a_i,a_j)+\gamma(a_i,a_k))Z\cdot N(0,\sigma^2)}\right]\mathbb{E}\left[e^{Z\cdot N(0,2\sigma^2)}\right]-e^{2(Z\cdot\sigma)^2}\\
&\leqslant\mathbb{E}\left[e^{2Z\cdot N(0,\sigma^2)}\right]\mathbb{E}\left[e^{Z\cdot N(0,2\sigma^2)}\right]-e^{2(Z\cdot\sigma)^2}\\
&=e^{3(Z\cdot\sigma)^2}-e^{2(Z\cdot\sigma)^2}.
\end{aligned}$$

Thus, we get that

$$\begin{aligned}
\mathsf{var}_{\mathbf{u}}(\Psi(a,\mathbf{u}))&\leqslant\sum_{i<j}e^{-2Z\cdot d(a_i,a_j)}(e^{4(Z\cdot\sigma)^2}-e^{2(Z\cdot\sigma)^2})+\sum_{(i,j,k)\text{ distinct}}e^{-Z\cdot(d(a_i,a_j)+d(a_i,a_k))}(e^{3(Z\cdot\sigma)^2}-e^{2(Z\cdot\sigma)^2})\\
&\leqslant(e^{4(Z\cdot\sigma)^2}-e^{2(Z\cdot\sigma)^2})\sum_i\left(\sum_{j\neq i}e^{-2Z\cdot d(a_i,a_j)}+\sum_{j,k\neq i}e^{-Z\cdot(d(a_i,a_j)+d(a_i,a_k))}\right)\\
&=(e^{4(Z\cdot\sigma)^2}-e^{2(Z\cdot\sigma)^2})\sum_i\left(\sum_{j\neq i}e^{-2Z\cdot d(a_i,a_j)}\right)^2\\
&=(e^{4(Z\cdot\sigma)^2}-e^{2(Z\cdot\sigma)^2})\cdot\sum_i C_i^2.
\end{aligned}$$

Therefore, using Claim 17 we conclude that

$$\mathsf{var}_{\mathbf{u}}(\Psi(a,\mathbf{u}))\leqslant(e^{4(Z\cdot\sigma)^2}-e^{2(Z\cdot\sigma)^2})\frac{c_3\Psi(a)}{\log n}\cdot\sum_i C_i=\frac{2c_3}{\log n}\cdot(e^{4(Z\cdot\sigma)^2}-e^{2(Z\cdot\sigma)^2})\cdot\Psi(a)^2.$$

Setting $c_1:=2c_3$ completes the proof. □

Putting the last two claims together, we have:

Claim 19. *Let $\sigma=10^4\sqrt{c_1}\frac{\sqrt{\log n}}{n}$ and let $a\in\mathbb{R}^n$ be good. Then*

$$\Pr_{\mathbf{u}}[\Psi(a,\mathbf{u})>\Psi(a)+\frac{(Z\sigma)^4}{2}\Psi(a)]\geqslant 0.96.$$

Proof. We upper bound the probability of the complement event. Using Claim 15 (and $e^t\geqslant 1+t+t^2/2$ for $t\geqslant 0$), we get

$$\mathop{\mathbb{E}}_{\mathbf{u}}[\Psi(a,\mathbf{u})]\geqslant\Psi(a)\cdot\left(1+(Z\sigma)^2+\frac{(Z\sigma)^4}{2}\right).$$

Hence

$$\Pr_{\mathbf{u}}\left[\Psi(a,\mathbf{u})\leqslant\Psi(a)+\frac{(Z\sigma)^4}{2}\Psi(a)\right]\leqslant\Pr_{\mathbf{u}}\left[\left|\Psi(a,\mathbf{u})-\mathop{\mathbb{E}}_{\mathbf{u}'}[\Psi(a,\mathbf{u}')]\right|\geqslant\Psi(a)\cdot(Z\sigma)^2\right].$$

We want to upper bound the probability of the last event using Chebyshev's inequality. Since a is good, the conclusion of Claim 18 holds. Since $Z\sigma=o(1)$, for large enough n we get

$$\mathsf{var}_{\mathbf{u}}[\Psi(a,\mathbf{u})]\leqslant\frac{c_1}{\log n}\cdot(e^{4(Z\cdot\sigma)^2}-e^{2(Z\cdot\sigma)^2})\cdot\Psi(a)^2\leqslant\frac{c_1}{\log n}\cdot\Psi(a)^2\cdot 8(Z\sigma)^2.$$

Therefore, applying Chebyshev's inequality we see the probability in question is at most

$$\frac{\mathsf{var}_{\mathbf{u}}[\Psi(a,\mathbf{u})]}{\Psi(a)^2\cdot(Z\sigma)^4} \leqslant \frac{c_1\cdot\Psi(a)^2\cdot 8(Z\sigma)^2}{(\log n)\cdot\Psi(a)^2\cdot(Z\sigma)^4} = \frac{8c_1}{(\log n)\cdot(Z\sigma)^2} = \frac{4c_1}{10^2 c_1} = 0.04. \quad \square$$

3.3 Finishing the argument

For each u, denote $\varepsilon_u = \Pr_{\mathbf{a}\in D}[\ell_{\mathbf{a},u} \not\subseteq D]$, and denote $\varepsilon = \mathbb{E}_{\mathbf{u}\sim\mathcal{N}(0,\sigma^2 I_n)}[\varepsilon_{\mathbf{u}}] = \Pr_{\mathbf{a},\mathbf{u}}[\ell_{\mathbf{a},\mathbf{u}} \not\subseteq D]$.

Claim 20. *For each u, $\mathcal{D}_{TV}[\mathbf{a};\mathbf{a}-u] \leqslant \varepsilon_u + \varepsilon_{-u}$.*

Proof. Let K be a Borel set. Note that it is enough to show that (1) if $K \subseteq D$ then $0 \leqslant \Pr_{\mathbf{a}\in D}[\mathbf{a}\in K] - \Pr_{\mathbf{a}\in D}[\mathbf{a}-u\in K] \leqslant \varepsilon_u$, and (2) if $K \subseteq \bar{D}$, then $-\varepsilon_{-u} \leqslant \Pr_{\mathbf{a}\in D}[\mathbf{a}\in K] - \Pr_{\mathbf{a}\in D}[\mathbf{a}-u\in K] \leqslant 0$. Indeed, given both (1) and (2), the triangle inequality implies for any Borel set $K\subseteq\mathbb{R}^n$,

$$\left|\Pr_{\mathbf{a}\in D}[\mathbf{a}\in K] - \Pr_{\mathbf{a}\in D}[\mathbf{a}-u\in K]\right|$$
$$\leqslant \left|\Pr_{\mathbf{a}\in D}[\mathbf{a}\in K\cap D] - \Pr_{\mathbf{a}\in D}[\mathbf{a}-u\in K\cap D] + \Pr_{\mathbf{a}\in D}[\mathbf{a}\in K\setminus D] - \Pr_{\mathbf{a}\in D}[\mathbf{a}-u\in K\setminus D]\right| \leqslant \varepsilon_u+\varepsilon_{-u}.$$

To prove (1), note that $\Pr_{\mathbf{a}\in D}[\mathbf{a}\in K] = \mu(K)$ and

$$\Pr_{\mathbf{a}\in D}[\mathbf{a}-u\in K] = \Pr_{\mathbf{a}\in D}[\mathbf{a}\in K+u] = \mu((K+u)\cap D).$$

This is at most $\mu(K+u)=\mu(K)$ (hence the expression in (1) is non-negative) and at least $\geqslant \mu(K+u)-\mu((K+u)\setminus D) = \mu(K)-\mu(K\setminus(D-u))$. Therefore

$$0 \leqslant \Pr_{\mathbf{a}\in D}[\mathbf{a}\in K] - \Pr_{\mathbf{a}\in D}[\mathbf{a}-u\in K] \leqslant \mu(K\setminus(D-u)) \leqslant \mu(D\setminus(D-u)) = \Pr_{\mathbf{a}\in D}[\mathbf{a}+u\notin D] \leqslant \varepsilon_u.$$

To prove (2), note that $\Pr_{\mathbf{a}\in D}[\mathbf{a}\in K]=0$ (hence the expression in (2) is non-positive) and

$$\Pr_{\mathbf{a}\in D}[\mathbf{a}-u\in K] \leqslant \Pr_{\mathbf{a}\in D}[\mathbf{a}-u\notin D] \leqslant \varepsilon_{-u}. \qquad \square$$

Claim 21. $\varepsilon \geqslant 0.1$.

Proof. Let E_1 be the event that $\mathbf{a}+\mathbf{u}[0,1]\subseteq D$, let E_2 be the event that $\Psi(\mathbf{a}) \leqslant 1$, let E_3 be the event that $|\mathbf{u}_i| > 1/20$ for some i and let E_4 be the event that $\Psi(\mathbf{a},\mathbf{u}) > \Psi(\mathbf{a}) + \frac{(Z\sigma)^4}{2}\Psi(\mathbf{a})$. Finally, let E_5 be the event that $\Psi(\mathbf{a}+\mathbf{u}) > \Psi(\mathbf{a})$ and denote $E(\mathbf{a},\mathbf{u}) = E_1\cap(\neg E_2)\cap(\neg E_3)\cap E_4$. Note that if the event E holds for a,u, then E_5 also holds, since by Claim 14:

$$\Psi(a+u) \geqslant \Psi(a,u) - n^2\cdot e^{-Z/4} > \Psi(a) + \frac{(Z\sigma)^4}{2}\Psi(a) - n^2\cdot e^{-Z/4} \geqslant \Psi(a).$$

By Claim 16 the probability of E_2 is at most the probability $\mathbf{a}$ is bad, hence it is at most 0.005. By definition, the probability of E_1 is $1-\varepsilon$. By the union bound and Chernoff inequality, the probability of E_3 is $o(1)$. Thus, by Claim 19 we have

$$\Pr_{\mathbf{a},\mathbf{u}}[E(\mathbf{a},\mathbf{u})] \geqslant 0.96 - \varepsilon - 0.005 - o(1) \geqslant 0.95 - \varepsilon. \tag{1}$$

Fix u. Using Claim 20 we get that

$$\Pr_{\mathbf{a}}[E(\mathbf{a}-u,u)] \geqslant \Pr_{\mathbf{a}}[E(\mathbf{a},u)] - \mathcal{D}_{TV}[\mathbf{a};\mathbf{a}-u] \geqslant \Pr_{\mathbf{a}}[E(\mathbf{a},u)] - \varepsilon_u - \varepsilon_{-u}.$$

By the union bound, we now conclude that

$$\Pr_{\mathbf{a}}[E(\mathbf{a}-u,u) \cap E(\mathbf{a},u)] \geqslant 1 - \Pr_{\mathbf{a}}\left[\overline{E(\mathbf{a}-u,u)}\right] - \Pr_{\mathbf{a}}\left[\overline{E(\mathbf{a},u)}\right] \geqslant 2\Pr_{\mathbf{a}}[E(\mathbf{a},u)] - 1 - \varepsilon_u - \varepsilon_{-u}.$$

Taking expectation over $\mathbf{u}$, we get that

$$\Pr_{\mathbf{a},\mathbf{u}}[E(\mathbf{a}-\mathbf{u},\mathbf{u}) \cap E(\mathbf{a},\mathbf{u})] \geqslant 2\Pr_{\mathbf{a},\mathbf{u}}[E(\mathbf{a},\mathbf{u})] - 1 - 2\mathop{\mathbb{E}}_{\mathbf{u}}[\varepsilon_{\mathbf{u}}] \geqslant 0.9 - 4\varepsilon.$$

Next, when both $E(a-u,u)$ and $E(a,u)$ hold, we have by the previous observation that E_5 holds for both pairs $(a-u,u)$ and (a,u), and so $\Psi(a+u) > \Psi(a) = \Psi((a-u)+u) > \Psi(a-u)$. Thus, we get that $\Pr_{\mathbf{a},\mathbf{u}}[\Psi(\mathbf{a}+\mathbf{u}) > \Psi(\mathbf{a}-\mathbf{u})] \geqslant 0.9 - 4\varepsilon$. On the other hand, the probability on the left hand side is at most 0.5; this follows as $\Pr_{\mathbf{a},\mathbf{u}}[\Psi(\mathbf{a}+\mathbf{u}) > \Psi(\mathbf{a}-\mathbf{u})] = \Pr_{\mathbf{a},\mathbf{u}}[\Psi(\mathbf{a}-\mathbf{u}) > \Psi(\mathbf{a}+\mathbf{u})]$ (since the distributions of $\mathbf{u}$ and $-\mathbf{u}$ are identical) and their sum is at most 1. Combining the two inequalities we get that $\varepsilon \geqslant 0.1$. □

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